

Data Analysis — Cocktail Recipes MVP

1. Project Description

This document describes the data analysis process for the Cocktail Recipes MVP website.

The goal of the analysis:

- verify event tracking
- analyze user behavior
- analyze search and filters usage
- identify popular cocktails
- build product metrics
- create dashboard
- generate product insights

Data source:

Google BigQuery

Tables used:

cocktails.events
cocktails.sessions
cocktails.page_views
cocktails.cocktails
cocktails.ingredients
cocktails.cocktail_ingredients
cocktails.tags
cocktails.cocktail_tags
cocktails.cocktail_similar

2. Checking Tracking Implementation

2.1 Check if tables exist

SQL:

```
SELECT *  
FROM cocktail-analytics.cocktails.event  
LIMIT 10
```

Untitled query

Run

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1 SELECT *

2 FROM cocktail-analytics.cocktails.event

3 LIMIT 10

Query completed

Query results

Create conversationSave resultsOpen in

Job informationResultsVisualizationJSONExecution detailsExecution graph

Row	id	event_name	timestamp	anonymous_user_id	page_name	page_url	search_text	filters_applied	filter_type	filter_value
1	665	page_view	2026-04-06T08:36:48.733729	2192bf06-d80c-4246-923e-089...	home	http://cocktail-catalogue.onrender.com/api/cocktails?min_price=0&max_price=180&page=1&page_size=12	null	null	null	null
2	666	cocktail_card_view	2026-04-06T08:36:48.791484	2192bf06-d80c-4246-923e-089...	null	null	null	null	null	null
3	667	cocktail_card_view	2026-04-06T08:36:48.791484	2192bf06-d80c-4246-923e-089...	null	null	null	null	null	null
4	668	cocktail_card_view	2026-04-06T08:36:48.791484	2192bf06-d80c-4246-923e-089...	null	null	null	null	null	null
5	669	page_view	2026-04-06T08:36:53.498032	affc9b86-c8ae-49a2-8304-ab5d...	home	http://cocktail-catalogue.onrender.com/api/cocktails?min_price=0&max_price=180&page=1&page_size=12	null	null	null	null
6	670	cocktail_card_view	2026-04-06T08:36:53.531660	affc9b86-c8ae-49a2-8304-ab5d...	null	null	null	null	null	null
7	671	cocktail_card_view	2026-04-06T08:36:53.531660	affc9b86-c8ae-49a2-8304-ab5d...	null	null	null	null	null	null
8	672	cocktail_card_view	2026-04-06T08:36:53.531660	affc9b86-c8ae-49a2-8304-ab5d...	null	null	null	null	null	null
9	673	page_view	2026-04-06T18:49:30.682311	7d3b4357-0717-40f1-8102-53b...	home	http://cocktail-catalogue.onrender.com/api/cocktails?min_price=0&max_price=180&page=1&page_size=12	null	null	null	null
10	674	cocktail_card_view	2026-04-06T18:49:30.921077	7d3b4357-0717-40f1-8102-53b...	null	null	null	null	null	null

Results per page: 501 - 10 of 10

SELECT *
FROM cocktail-analytics.cocktails.session
LIMIT 10

Untitled query

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1 SELECT *

2 FROM cocktail-analytics.cocktails.session

3 LIMIT 10

Query completed

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Query results

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Job informationResultsVisualizationJSONExecution detailsExecution graph

Row	id	anonymous_user_id	session_start	session_end	device_type	browser
1	166	2192bf06-d80c-4246-923e-089...	2026-04-06T08:36:48.733729	2026-04-06T08:36:48.733729	Windows	Chrome
2	167	affc9b86-c8ae-49a2-8304-ab5d...	2026-04-06T08:36:53.498032	2026-04-06T08:36:53.498032	Windows	Chrome
3	168	7d3b4357-0717-40f1-8102-53b...	2026-04-06T18:49:30.682311	2026-04-06T18:49:30.682311	Android	Chrome Mobile
4	169	a87af6a8-ea78-4cd1-b360-e03a...	2026-04-06T18:49:34.203468	2026-04-06T18:49:34.203468	Android	Chrome Mobile
5	170	02fa8bfc-9419-45e0-a545-231a...	2026-04-06T18:49:39.039075	2026-04-06T18:49:50.674300	Android	Chrome Mobile
6	171	bfb78644-6c76-42d9-9816-14f3...	2026-04-06T18:49:57.288391	2026-04-06T18:49:57.288391	Android	Chrome Mobile
7	172	5354d74b-3e4c-40eb-83be-125...	2026-04-06T18:50:00.304694	2026-04-06T18:50:00.304694	Android	Chrome Mobile
8	173	4376f7de-1df1-4786-b104-c8d0...	2026-04-06T18:50:00.611941	2026-04-06T18:50:00.611941	Android	Chrome Mobile
9	174	293b0047-6702-41bd-89a2-5e8...	2026-04-06T18:50:01.185800	2026-04-06T18:50:01.185800	Android	Chrome Mobile
10	175	3b29a0f6-f2ea-40fe-9199-aff17...	2026-04-06T18:50:01.695765	2026-04-06T18:50:01.695765	Android	Chrome Mobile

2.2 Check events structure

SQL:

```
SELECT column_name, data_type
FROM cocktail-analytics.cocktails.INFORMATION_SCHEMA.COLUMNS
WHERE table_name = 'event'
```

Screenshot:

Untitled query

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```
1 SELECT column_name, data_type
2 FROM cocktail-analytics.cocktails.INFORMATION_SCHEMA.COLUMNS
3 WHERE table_name = 'event'
4
```

Query completed

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Query results

Job information

Results

Visualization

JSON

Execution details

Row	column_name	data_type	
1	id	INT64	
2	event_name	STRING	
3	timestamp	DATETIME	
4	anonymous_user_id	STRING	
5	page_name	STRING	
6	page_url	STRING	
7	search_text	STRING	
8	filters_applied	BOOL	
9	filter_type	STRING	
10	filter_value	STRING	
11	previous_filters	STRING	
12	results_count	INT64	
13	servings_number	INT64	
14	age_confirmed	BOOL	
15	position	INT64	
16	cocktail_id	INT64	
17	session_id	INT64	

Expected fields:

1. event_id
 2. event_name
 3. timestamp
 4. anonymous_user_id
 5. page_name
 6. session_id
 7. cocktail_id
 8. search_text
 9. filter_type
 10. filter_value
 11. previous_filters
 12. filters_applied
 13. results_count
 14. servings_number
 15. age_confirmed
 16. position
 17. page_url
-

2.3 Check events recording

SQL:

```
SELECT
event_name,
COUNT(*) count
FROM `cocktail-analytics.cocktails.event`
GROUP BY event_name
ORDER BY count DESC
```

Screenshot:

Untitled query

RunDownloadShareScheduleOpen in

```
1 SELECT
2 event_name,
3 COUNT(*) count
4 FROM `cocktail-analytics.cocktails.event`
5 GROUP BY event_name
6 ORDER BY count DESC
7
8
```

Query completed

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Query results

Job informationResultsVisualizationJSONExecution detailsExecution graph

Row	event_name	count
1	cocktail_card_view	784
2	page_view	536
3	servings_changed	384
4	filter_applied	216
5	cocktail_page_open	131
6	filters_reset	36
7	search_query	24
8	age_confirmation	9

3. Base Analytics

3.1 Search Analysis

SQL:

```
SELECT
search_text,
COUNT(*) searches
FROM `cocktail-analytics.cocktails.event`
WHERE event_name = 'search_query'
GROUP BY search_text
ORDER BY searches DESC
```

Screenshot:

Untitled query

Run

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```
1 SELECT
2 search_text,
3 COUNT(*) searches
4 FROM `cocktail-analytics.cocktails.event`
5 WHERE event_name = 'search_query'
6 GROUP BY search_text
7 ORDER BY searches DESC
8
9
```

Query completed

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Query results

Create conversation

Save results

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Job information

Results

Visualization

JSON

Execution details

Execution graph

Row	search_text	searches
1	Mojito	5
2	mojito	3
3	When well made, this Cuban co...	2
4	br	1
5	gijkl	1
6	ios, this br	1
7	brown	1
8	brown	1
9	opac	1
10	d	1
11	Due to its proximity to the movi...	1
12	tpg	1
13	fd	1
14	Mojoti	1
15	Mojito\	1
16	Mojito	1
17	moj	1

Results per page: 50 1 - 17 of 17

3.2 Search with no results

SQL:

```
SELECT
search_text,
COUNT(*) AS searches
FROM `cocktail-analytics.cocktails.event`
WHERE event_name = 'search_query'
AND results_count = 0
GROUP BY search_text
ORDER BY searches DESC
```

Screenshot:

Untitled query

Run

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```

1 SELECT
2 search_text,
3 COUNT(*) AS searches
4 FROM `cocktail-analytics.cocktails.event`
5 WHERE event_name = 'search_query'
6 AND results_count > 0
7 GROUP BY search_text
8 ORDER BY searches DESC
9
10

```

Query completed

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Query results

Create conversation

Save results

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Job information

Results

Visualization

JSON

Execution details

Execution graph

Row	search_text	searches
1	Mojito	1
2	gfkj	1
3	fd	1
4	npp	1
5	Mojito	1
6	opon	1
7	Mojoti	1

Results per page: 50 1 - 7 of 7 |< < > >|

3.3 Cocktail popularity

SQL:

```
SELECT
c.name,
COUNT(*) AS views
FROM `cocktail-analytics.cocktails.event` e
JOIN `cocktail-analytics.cocktails.cocktail` c
ON e.cocktail_id = c.id
WHERE e.event_name = 'cocktail_page_open'
GROUP BY c.name
ORDER BY views DESC
```

Screenshot:

Untitled query
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```

1 SELECT
2   c.name,
3   COUNT(*) AS views
4 FROM `cocktail-analytics.cocktails.event` e
5 JOIN `cocktail-analytics.cocktails.cocktail` c
6   ON e.cocktail_id = c.id
7 WHERE e.event_name = 'cocktail_page_open'
8 GROUP BY c.name
9 ORDER BY views DESC
10
11
12

```

This query will process 43 KB when run.
 Using on-demand processing quota

Query results

Job information

Results

Visualization

JSON

Execution details

Execution graph

Row	name	views
1	Brown Derby	68
2	White Rob Roy	33
3	Mojito	30

3.4 Filter analysis

SQL:

```

SELECT
  filter_type,
  TRIM(value) AS filter_value,
  COUNT(*) AS count
FROM `cocktail-analytics.cocktails.event`,
  UNNEST(SPLIT(filter_value, ',')) AS value
WHERE event_name = 'filter_applied'
GROUP BY 1,2
ORDER BY count DESC

```

Screenshot:

Untitled query
Run
Download
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Open in >
More >
Save >

```

1 SELECT
2   filter_type,
3   TRIM(value) AS filter_value,
4   COUNT(*) AS count
5 FROM `cocktail-analytics.cocktails_event`
6 UNNEST(SPLIT(filter_value, ',')) AS value
7 WHERE event_name = 'filter_applied'
8 GROUP BY 1,2
9 ORDER BY count DESC
10
11
12

```

Query completed

Using on-demand processing quota

Query results

[+ Create conversation](#)
[Save results >](#)
[Open in >](#)

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	filter_type	filter_value	count		
1	ingredients	whisky	37		
2	ingredients	vermouth	36		
3	ingredients	bourbon	23		
4	vibes	romantic	18		
5	vibes	party	15		
6	vibes	classic	15		
7	alcohol_level	medium	12		
8	ingredients	vodka	11		
9	ingredients	rum	10		
10	vibes	beach	7		
11	vibes	tropical	6		
12	sweetness_level	medium	5		
13	ingredients	beer	5		
14	min_price	33	4		
15	sweetness_level	dry	4		
16	vibes	summer	3		
17	sweetness_level	sweet	3		

Results per page: 50 | 1 - 33 of 33

Untitled query

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```

1 SELECT
2   i.name as ingredient_name,
3   COUNT(*) AS views
4 FROM `cocktail-analytics.cocktails_dev.event` e
5 JOIN `cocktail-analytics.cocktails_dev.cocktailingredients` ci
6   ON e.cocktail_id = ci.cocktail_id
7 JOIN `cocktail-analytics.cocktails_dev.ingredient` i
8   ON ci.ingredient_id = i.id
9 WHERE e.event_name = 'cocktail_page_open'
10 GROUP BY i.name
11 ORDER BY views DESC
12
13
14
15

```

Query completed
 Using on-demand processing quota

Query results

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	ingredient_name	views			
1	vermouth	1			
2	zeltser	1			
3	cherry	1			
4	whisky	1			

3.6 Auth events

```

SELECT
event_name,
COUNT(*) AS count
FROM `cocktail-analytics.cocktails.events`
WHERE event_name IN ('signup','login','logout')
GROUP BY event_name;

```

3.7 Authorized vs Anonymous

```

SELECT
CASE
WHEN user_id IS NULL THEN 'anonymous'
ELSE 'authorized'
END AS user_type,
COUNT(*) AS events
FROM `cocktail-analytics.cocktails.events`
GROUP BY user_type

```

3.8 Page → Rating conversion

```
SELECT
COUNT(DISTINCT e1.session_id) AS page_open_sessions,
COUNT(DISTINCT e2.session_id) AS rating_sessions,
COUNT(DISTINCT e2.session_id) / COUNT(DISTINCT e1.session_id) AS conversion
FROM `cocktail-analytics.cocktails.events` e1
LEFT JOIN `cocktail-analytics.cocktails.events` e2
ON e1.session_id = e2.session_id
AND e2.event_name = 'rating_added'
WHERE e1.event_name = 'cocktail_page_open'
```

We take all sessions with cocktail_page_open and check whether a rating_added event occurred in the same session. Then we calculate conversion as the ratio of such sessions.

4. Product Metrics

4.1 Sessions per day

SQL:

```
SELECT
DATE(session_start) date,
COUNT(*) sessions
FROM `cocktail-analytics.cocktails.session`
GROUP BY date
ORDER BY date
```

Untitled query

Run

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```

1 SELECT
2 DATE(session_start) date,
3 COUNT(*) sessions
4 FROM `cocktail-analytics.cocktails.session`
5 GROUP BY date
6 ORDER BY date
7
8
9

```

Query completed

Using on-demand processing quota

Query results

Job information

Results

Visualization

JSON

Execution details

Execution graph

Row	date	sessions
1	2026-03-25	7
2	2026-03-26	86
3	2026-03-27	433
4	2026-03-28	2
5	2026-03-29	20
6	2026-03-30	82
7	2026-03-31	12
8	2026-04-01	21
9	2026-04-02	3
10	2026-04-03	12
11	2026-04-04	13
12	2026-04-06	12

4.2 Events per session

SQL:

```

SELECT
session_id,
COUNT(*) events
FROM `cocktail-analytics.cocktails.event`
GROUP BY session_id

```

Untitled query
 Run
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```

1 SELECT
2   session_id,
3   COUNT(*) events
4 FROM `cocktail-analytics.cocktails.event`
5 GROUP BY session_id
6
7
8

```

✓ Query completed
Using on-demand processing quota

Query results

Job information	Results	Visualization	JSON	Execution detail
Row	session_id	events		
1	339	1		
2	105	8		
3	109	10		
4	110	11		
5	111	10		
6	112	8		
7	285	4		
8	286	4		
9	287	4		
10	289	4		
11	291	4		
12	298	4		

4.3 Cocktail CTR (Card → Page conversion per cocktail)

This metric shows how often users open a cocktail page after seeing the cocktail card.

CTR helps identify:

- attractive cocktails
- low-quality search ranking
- position bias
- hidden popular cocktails

Formula:

CTR = cocktail_page_open sessions / cocktail_card_view sessions

```
SELECT
c.name,
e.cocktail_id,
COUNT(DISTINCT IF(e.event_name = 'cocktail_page_open',
e.session_id, NULL))/
NULLIF(COUNT(DISTINCT IF(
e.event_name = 'cocktail_card_view',
e.session_id,
NULL)),0) AS ctr

FROM `cocktail-analytics.cocktails.event` e

JOIN `cocktail-analytics.cocktails.cocktail` c
ON e.cocktail_id = c.id

WHERE e.event_name IN (
'cocktail_card_view',
'cocktail_page_open'
)

GROUP BY
c.name,
e.cocktail_id

ORDER BY ctr DESC
```

Screenshot:

Untitled query
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```

1 SELECT
2   c.name,
3   e.cocktail_id,
4   COUNT(DISTINCT IF(e.event_name = 'cocktail_page_open',
5     e.session_id,NULL))/
6   NULLIF(COUNT(DISTINCT IF(
7     e.event_name = 'cocktail_card_view',
8     e.session_id,
9     NULL)),0) AS ctr
10
11
12 FROM `cocktail-analytics.cocktails.event` e
13
14
15 JOIN `cocktail-analytics.cocktails.cocktail` c
16 ON e.cocktail_id = c.id
17
18
19 WHERE e.event_name IN (
20   'cocktail_card_view',
21   'cocktail_page_open'
22 )
23
24
25 GROUP BY
26   c.name,
27   e.cocktail_id
28
29
30 ORDER BY ctr DESC
31
32

```

✓ This query will process 59.56 KB when run.

Using on-demand processing quota

Query results

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	name	cocktail_id	ctr		
1	Brown Derby	3	0.30555555555555...		
2	Mojito	2	0.149253731343...		
3	White Rob Roy	1	0.140845070422...		

4.4 Calculator usage

```

SELECT
COUNTIF(event_name = 'servings_changed') / NULLIF(COUNTIF(event_name =
'cocktail_page_open'),0) AS calculator_usage
FROM `cocktail-analytics.cocktails.event`

```

Untitled query

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```

1 SELECT
2 COUNTIF(event_name = 'servings_changed') / NULLIF(COUNTIF(event_name = 'cocktail_page_open'),0) AS calculator_usage
3 FROM `cocktail-analytics.cocktails.event`
4
5

```

Query completed

Using on-demand processing quota

Query results

Job information
 Results
 Visualization
 JSON
 Execution details
 Execution graph

Row	calculator_usage
1	2.931297709923...

4.5 Search → View conversion

```

WITH s AS (
SELECT DISTINCT session_id
FROM `cocktail-analytics.cocktails.event`
WHERE event_name = 'search_query'
),

v AS (
SELECT DISTINCT session_id
FROM `cocktail-analytics.cocktails.event`
WHERE event_name = 'cocktail_page_open'
)

SELECT
COUNT(v.session_id) / COUNT(s.session_id)
AS conversion
FROM s
LEFT JOIN v USING(session_id)

```


Untitled query

Run

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```

1 WITH s AS (
2   SELECT DISTINCT session_id
3   FROM `cocktail-analytics.cocktails.event`
4   WHERE event_name = 'search_query'
5 )
6
7
8 v AS (
9   SELECT DISTINCT session_id
10  FROM `cocktail-analytics.cocktails.event`
11  WHERE event_name = 'cocktail_page_open'
12 )
13
14
15 SELECT
16   COUNT(v.session_id) / COUNT(s.session_id)
17   AS conversion
18 FROM s
19 LEFT JOIN v USING(session_id)
20
21

```

This query will process 51.33 KB when run.

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Query results

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	conversion				
1	0.2				

5. Dashboard

Dashboard Overview

Tool: Tableau

Data source: Google BigQuery

Goal:

Monitor user behavior
 Analyze search usage
 Analyze filters usage
 Analyze cocktail popularity
 Track product metrics
 Find product insights

SHEET 1 — Search & User Behavior

Goal:

- search usage
- zero results
- filters
- activity

```
SELECT
DATE(timestamp) AS date,
session_id,
search_text,
results_count,
filters_applied
FROM `cocktail-analytics.cocktails.event`
WHERE event_name = 'search_query'
```

✓ Chart 1 — Searches per day (Line)

date

COUNT(search_text)

✓ shows activity

✓ Chart 2 — Top searches (Bar)

search_text

COUNT(search_text)

Top 10

✓ shows popular requests

✓ Chart 3 — Searches with no results (Bar)

Filter:

results_count = 0

Fields:

search_text
COUNT

✓ shows problems

✓ Chart 4 — Searches with filters (Bar)

filters_applied
COUNT

✓ shows behavior

SHEET 2 — Cocktail Popularity & Product Metrics

Goal:

- popularity
 - CTR
 - ingredients
 - feature usage
-

✓ Query 2

SELECT

```
e.session_id,  
e.event_name,  
e.cocktail_id,  
c.name as cocktail_name,  
i.name as ingredient_name,  
e.servings_number
```

```
FROM `cocktail-analytics.cocktails.event` e
```

```
LEFT JOIN `cocktail-analytics.cocktails.cocktail` c  
ON e.cocktail_id = c.id
```

```
LEFT JOIN `cocktail-analytics.cocktails.cocktailingredients` ci  
ON e.cocktail_id = ci.cocktail_id
```

```
LEFT JOIN `cocktail-analytics.cocktails_dev.ingredient` i
ON ci.ingredient_id = i.id
```

✓ Chart 5 — Most viewed cocktails (Bar)

Filter:

event_name = cocktail_page_open

Fields:

cocktail_name
COUNT(*)

✓ popularity

✓ Chart 6 — Cocktail CTR (Scatter)

X:

COUNT(cocktail_card_view)

Y:

COUNT(cocktail_page_open) / COUNT(cocktail_card_view)

Label:

name

✓ Chart 7 — Popular ingredients (Bar)

Filter:

event_name = cocktail_page_open

Fields:

ingredient_name
COUNT(*)

✓ показыве join

✓ Chart 8 — Calculator usage

Filter:

event_name IN (
cocktail_page_open,
servings_changed
)

Metric:

COUNT(servings_changed) /COUNT(cocktail_page_open)

✓ feature usage
✓ product metric

6. Product Insights

Product Insights These are the things we can "read between the lines" for future growth:

- **Search Efficiency:** We can track top searches vs. "no results" queries. For example, if we see "Mojito" in both tables but with different spellings, that's an insight: users make typos. The system needs to be prepared for this.
- **Filter Usage:** We can see how often filters are used. This helps us understand if the interface is intuitive or if users even realize they can filter by ingredients or strength etc.
- **Content Popularity:** We can track the most viewed cocktails and ingredients. For instance, if we see high interest in bourbon or maple syrup, it's a clear hint on which recipes to add next.
- **Feature Engagement:** We track how users interact with specific tools, like the servings calculator. If we see a high engagement rate (e.g., 25%), it's a signal that this feature is valuable and should be improved.

7. Product Recommendations

Based on this data, even at the MVP stage, we can already plan our next steps:

1. **Improve Search Experience:** Implement "fuzzy search" and autocomplete to handle typos (like "Mojiti") and reduce "no results" pages.
2. **Content Optimization:** If people search for ingredients or cocktails that aren't in our database yet, it's a signal to add that content or improve localization.
3. **Farther Partner Opportunities:** As we are planning to expand website features and partnership with companies, we can provide alcohol brands with reports on how often their products are searched for or used in recipes. This helps them decide whether to advertise on our platform.